

Disproof of Conjecture 00000009689

Algebraic auxiliary values cannot add transcendence degree

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Claim addressed. The conjecture asserts that finitely many values $e^{\pi\sqrt{d_i}}$, together with the algebraic values of the modular j function at the corresponding CM points, have joint transcendence degree equal to the number of d_i plus one. Its singleton case would therefore have transcendence degree 2. This contradicts an elementary field-theoretic bound, irrespective of any independence or transcendence properties of the exponential values.

Proposition 1. *For every $x \in \mathbb{C}$ and every $j \in \mathbb{C}$ algebraic over $\mathbb{Q}$,*

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(x, j) \leq 1.$$

Proof. Put $K = \mathbb{Q}(j)$. Since j is algebraic, $K/\mathbb{Q}$ is algebraic and has transcendence degree zero. The algebra $K[x]$ is the image of the evaluation map $K[X] \rightarrow \mathbb{C}$ sending X to x . Thus it has transcendence degree at most 1 over K . Passing to its fraction field $K(x)$ preserves transcendence degree. The tower formula gives

$$\text{trdeg}_{\mathbb{Q}} K(x) = \text{trdeg}_{\mathbb{Q}} K + \text{trdeg}_K K(x) \leq 0 + 1 = 1.$$

Finally, $K(x) = \mathbb{Q}(x, j)$, proving the bound. □

Take the singleton $d_1 = 1$, a positive squarefree integer. Its exponential value is $x = e^{\pi}$. Let j_1 be the algebraic CM value specified by the conjecture. The proposition gives

$$\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(e^{\pi}, j_1) \leq 1 < 2,$$

whereas the conjecture prescribes $1 + 1 = 2$. This disproves the stated joint-degree clause and hence the conjecture as a conjunction. The same obstruction applies to every positive squarefree singleton d , so the argument would also work if one imposed $d > 1$.

Scope and the CM value. The source explicitly describes the auxiliary j values as algebraic. The proof uses exactly that property; it does not need to select a CM-point convention, compute a singular modulus, or define an alternative function called j . Even adjoining several algebraic CM values would not increase the transcendence degree. No assertion is made about the separate algebraic independence claim for distinct Gelfond values.

Lean correspondence and reproduction

The Lean theorem `adjoining_algebraic_value_bound` proves the proposition for all complex x, j with the actual Mathlib hypothesis `IsAlgebraic ℚ j`. It uses the genuine intermediate fields $\mathbb{Q}(j)$ and $\mathbb{Q}(j)(x)$, proves their iterated adjoining identity with $\mathbb{Q}(x, j)$, and applies the transcendence-degree tower formula. The one-generator bound is obtained from a surjective polynomial evaluation

map and the algebraic extension from a generated algebra to its fraction field. Both `Algebra.trdeg` and `IntermediateField.adjoin` are Mathlib’s actual definitions.

The definition `gelfondValue` uses the actual `Complex.exp`, `Real.pi`, and `Real.sqrt`. The formal theorem `gelfond_joint_degree_not_two` holds for every natural d and every algebraic auxiliary value. The final theorem says there is no assignment of algebraic auxiliary values to positive squarefree d satisfying the claimed degree 2 for all singleton cases. It explicitly instantiates $d = 1$ and verifies positivity and squarefreeness. Consequently the theorem covers any assignment of the algebraic CM values appearing in the source; it does not formalize or postulate a modular j function.

Inside `lean/`, run `lake build`, then

```
lake env lean -DwarningAsError=true Main.lean.
```

The project uses Lean 4.19.0 and Mathlib commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

The manifest pins the transitive dependencies. The proof is entirely algebraic; no numerical or experimental calculation is needed. The exact bilingual conjecture is preserved in `SOURCE.md`. Build logs, axiom output, source hashes, and PDF checks are in `verification/`. Author self-review and parent-agent adversarial review are recorded separately; no external independent review is claimed.

References

- [1] The Last Math Competition, Conjecture 00000009689 (original bilingual source).